

Hyperliquid Perpetual Futures — Deploy Spec

12 profitable strategies ranked. Per-strategy spec + capital + risk.

#	Strategy ID	Asset	TF	Hold	n	WR%	Sharpe	Total \$	\$/tr	Capital \$
1	D1_SOL_168h_z2.5_zwin60d_fund30davg	SOL	1h	168h	75	81.3	11.29	\$588	\$7.85	\$1,840
2	D1_SOL_168h_z2.0_zwin90d_zero_fund	SOL	1h	168h	80	77.5	12.02	\$719	\$8.99	\$1,963
3	D1_SOL_168h_z1.5_zwin90d_term_next	SOL	1h	168h	145	66.9	6.96	\$830	\$5.73	\$3,557
4	D1_SOL_72h_z1.5_zwin14d_term_next	SOL	1h	72h	259	56.8	4.66	\$874	\$3.37	\$2,723
5	D1_SOL_48h_z2.0_zwin7d_term_next	SOL	1h	48h	130	63.8	6.40	\$502	\$3.86	\$911
6	D1_ETH_24h_z1.5_zwin7d_fund30davg	ETH	1h	24h	296	56.1	3.27	\$519	\$1.75	\$1,037
7	A1_ETH_4h_Donchian_N50_ATR1.5	ETH	4h	Variable	249	47.0	3.32	\$147	\$0.59	\$250
8	A1_ETH_4h_Donchian_N20_ATR1.5	ETH	4h	Variable	427	45.2	3.16	\$110	\$0.26	\$250
9	A1_ETH_4h_Donchian_N50_ATR2.0	ETH	4h	Variable	204	42.6	1.99	\$94	\$0.46	\$250
10	C3_SOL_4h_vol_confirmed_Donchian	SOL	4h	Variable	211	50.7	5.28	\$1904	\$9.02	\$250
11	E4_Cyclops_confidence_sizing_SOL_4h	SOL	4h	Variable	134	56.7	1.98	\$286	\$2.13	\$250
12	F3_ETH_4h_rf_dist_bps_z1.5_momentum	ETH	4h	Variable	48	46.4	1.16	\$220	\$4.31	\$250

Notes:

- All metrics are from \$250 notional at 1x leverage in backtest. Scale linearly with notional. Capital column estimates working capital accounting for hold-overlap (multi-day holds may have several concurrent positions).
- Engine: Hyperliquid taker 4.5 bps x 2 + slippage 3 bps + hourly funding (1.25 bps/hr cap) + 50 ms latency. Same fees applied in every cell.
- T1 Carry (green) uses HL vs Binance basis. T2/T3/T4/T5 use Binance kline backbone (multi-year) for backtest; execute on HL.
- Sharpe is annualized. For T2-T5 the OOS Sharpe (out-of-sample test window) is shown when available.

#1 — D1_SOL_168h_z2.5_zwin60d_fund30davg

Tier: T1 Carry

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	1h
Hold horizon	168h (7 days)
Backtest window	HL 107d (Jan 30 - May 16 2026)

Signal & Exit

Field	Value
Entry trigger	basis_z > +2.5 -> LONG; basis_z < -2.5 -> SHORT
Expected-basis proxy	30d funding mean x (168h/8h)
z-score window	60 days
Exit rule	Time stop 168h (no ATR early exit)

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$1,840
Max drawdown (on \$250 notional)	as reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	75
Win rate	81.3%
Sharpe (annualized)	11.29
Total P&L (backtest)	\$588.41
Avg P&L per trade	\$7.85
Avg funding paid per trade	\$-0.130
Avg fee paid per trade	\$0.225
Permutation p-value	0.0010
Bootstrap 95% CI on \$/tr	[\$0.44, \$1.09]

Validation gates passed	7/7
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Caveats: Small effective n (75 trades / 107 days at 168h hold = limited statistical power). Re-validate when more HL data accumulates.

See "Basis Carry (D1) — Spot-Perp Mispricing Reversion" on the explanation pages.

#2 — D1_SOL_168h_z2.0_zwin90d_zero_fund

Tier: T1 Carry

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	1h
Hold horizon	168h (7 days)
Backtest window	HL 107d

Signal & Exit

Field	Value
Entry trigger	basis_z > +2.0 -> LONG; basis_z < -2.0 -> SHORT
Expected-basis proxy	zero (pure spot-perp, no funding adjustment)
z-score window	90 days
Exit rule	Time stop 168h

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$1,963
Max drawdown (on \$250 notional)	as reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	80
Win rate	77.5%
Sharpe (annualized)	12.02
Total P&L (backtest)	\$718.90
Avg P&L per trade	\$8.99
Avg funding paid per trade	\$-0.130
Avg fee paid per trade	\$0.225
Permutation p-value	0.0010
Bootstrap 95% CI on \$/tr	[\$0.49, \$1.12]

Validation gates passed	7/7
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Caveats: 168h hold = small effective n. Validate on extended history.

See "Basis Carry (D1) — Spot-Perp Mispricing Reversion" on the explanation pages.

#3 — D1_SOL_168h_z1.5_zwin90d_term_next

Tier: T1 Carry

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	1h
Hold horizon	168h (7 days)
Backtest window	HL 107d

Signal & Exit

Field	Value
Entry trigger	basis_z > +1.5 -> LONG; basis_z < -1.5 -> SHORT
Expected-basis proxy	current funding x (168h/8h) = 'term_next'
z-score window	90 days
Exit rule	Time stop 168h

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$3,557
Max drawdown (on \$250 notional)	as reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	145
Win rate	66.9%
Sharpe (annualized)	6.96
Total P&L (backtest)	\$830.19
Avg P&L per trade	\$5.73
Avg funding paid per trade	\$-0.110
Avg fee paid per trade	\$0.225
Permutation p-value	0.0010
Validation gates passed	7/7

Caveats: Looser $z=1.5$ = more trades + larger total PnL than higher- z variants. Best mid-Sharpe / large-n SOL 168h cell.

See "*Basis Carry (D1) — Spot-Perp Mispricing Reversion*" on the explanation pages.

#4 — D1_SOL_72h_z1.5_zwin14d_term_next

Tier: T1 Carry

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	1h
Hold horizon	72h (3 days)
Backtest window	HL 107d

Signal & Exit

Field	Value
Entry trigger	basis_z > +1.5 -> LONG; basis_z < -1.5 -> SHORT
Expected-basis proxy	current funding x 9 = 'term_next'
z-score window	14 days
Exit rule	Time stop 72h

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$2,723
Max drawdown (on \$250 notional)	as reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	259
Win rate	56.8%
Sharpe (annualized)	4.66
Total P&L (backtest)	\$873.66
Avg P&L per trade	\$3.37
Avg funding paid per trade	\$-0.110
Avg fee paid per trade	\$0.225
Permutation p-value	0.0010
Bootstrap 95% CI on \$/tr	[\$0.18, \$0.41]

Validation gates passed	7/7
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Caveats: Highest total PnL with manageable hold. Recommended SOL deploy.

See "Basis Carry (D1) — Spot-Perp Mispricing Reversion" on the explanation pages.

#5 — D1_SOL_48h_z2.0_zwin7d_term_next

Tier: T1 Carry

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	1h
Hold horizon	48h (2 days)
Backtest window	HL 107d

Signal & Exit

Field	Value
Entry trigger	basis_z > +2.0 -> LONG; basis_z < -2.0 -> SHORT
Expected-basis proxy	current funding x 6 = 'term_next'
z-score window	7 days
Exit rule	Time stop 48h

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$911
Max drawdown (on \$250 notional)	as reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	130
Win rate	63.8%
Sharpe (annualized)	6.40
Total P&L (backtest)	\$501.91
Avg P&L per trade	\$3.86
Avg funding paid per trade	\$-0.130
Avg fee paid per trade	\$0.225
Permutation p-value	0.0010
Bootstrap 95% CI on \$/tr	[\$0.24, \$0.57]

Validation gates passed	7/7
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Caveats: Stricter z threshold + shorter hold = good Sharpe with moderate n .

See "Basis Carry (D1) — Spot-Perp Mispricing Reversion" on the explanation pages.

#6 — D1_ETH_24h_z1.5_zwin7d_fund30davg

Tier: T1 Carry

Market & Execution

Field	Value
Asset	ETH
Venue	Hyperliquid Perp
Signal timeframe	1h
Hold horizon	24h (1 day)
Backtest window	HL 107d

Signal & Exit

Field	Value
Entry trigger	basis_z > +1.5 -> LONG; basis_z < -1.5 -> SHORT
Expected-basis proxy	30d funding mean x 3
z-score window	7 days
Exit rule	Time stop 24h

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$1,037
Max drawdown (on \$250 notional)	as reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	296
Win rate	56.1%
Sharpe (annualized)	3.27
Total P&L (backtest)	\$518.66
Avg P&L per trade	\$1.75
Avg funding paid per trade	\$-0.030
Avg fee paid per trade	\$0.225
Permutation p-value	0.0010
Bootstrap 95% CI on \$/tr	[\$0.10, \$0.32]

Validation gates passed	7/7
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Caveats: Best 24h-hold cell. Easiest to operate. RECOMMENDED FIRST DEPLOY.

See "Basis Carry (D1) — Spot-Perp Mispricing Reversion" on the explanation pages.

#7 — A1_ETH_4h_Donchian_N50_ATR1.5

Tier: T2 Trend

Market & Execution

Field	Value
Asset	ETH
Venue	Hyperliquid Perp
Signal timeframe	4h
Hold horizon	Variable (avg ~4.7 bars = ~19h)
Backtest window	~8.6y Binance ETH spot (2017-08 -> 2026-03)

Signal & Exit

Field	Value
Entry trigger	Close > prior 50-bar HIGH -> LONG; Close < prior 50-bar LOW -> SHORT
Exit rule	ATR(14) trailing stop, mult = 1.5

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$250
Max drawdown (on \$250 notional)	\$88.46
Max DD as % of notional	-35.4%
Profit factor	1.19
Calmar (PnL / MaxDD)	1.66

Performance Metrics

Metric	Value
Total trades (n)	249
Win rate	47.0%
Sharpe (annualized)	3.32 [OOS]
Total P&L (backtest)	\$146.67
Avg P&L per trade	\$0.59
Validation gates passed	OOS PASS + beats BH 5.6x

Caveats: Max DD only -\$88 on \$250 notional = -35% per-trade max DD. Beats ETH buy-and-hold Sharpe by 5.6x.

See "Donchian Channel Trend Following (A1)" on the explanation pages.

#8 — A1_ETH_4h_Donchian_N20_ATR1.5

Tier: T2 Trend

Market & Execution

Field	Value
Asset	ETH
Venue	Hyperliquid Perp
Signal timeframe	4h
Hold horizon	Variable (avg ~4.9 bars)
Backtest window	~8.6y Binance ETH

Signal & Exit

Field	Value
Entry trigger	Close > prior 20-bar HIGH -> LONG; Close < prior 20-bar LOW -> SHORT
Exit rule	ATR(14) trailing stop, mult = 1.5

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$250
Max drawdown (on \$250 notional)	\$108.31
Max DD as % of notional	-43.3%
Profit factor	1.09
Calmar (PnL / MaxDD)	1.02

Performance Metrics

Metric	Value
Total trades (n)	427
Win rate	45.2%
Sharpe (annualized)	3.16 [OOS]
Total P&L (backtest)	\$110.40
Avg P&L per trade	\$0.26
Validation gates passed	OOS PASS + beats BH

Caveats: Higher trade count than N=50 variant. Lower per-trade \$ but more samples.

See "Donchian Channel Trend Following (A1)" on the explanation pages.

#9 — A1_ETH_4h_Donchian_N50_ATR2.0

Tier: T2 Trend

Market & Execution

Field	Value
Asset	ETH
Venue	Hyperliquid Perp
Signal timeframe	4h
Hold horizon	Variable (avg ~7.9 bars = ~32h)
Backtest window	~8.6y Binance ETH

Signal & Exit

Field	Value
Entry trigger	Close > prior 50-bar HIGH -> LONG; Close < prior 50-bar LOW -> SHORT
Exit rule	ATR(14) trailing stop, mult = 2.0 (wider than rank 7)

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$250
Max drawdown (on \$250 notional)	\$97.52
Max DD as % of notional	-39.0%
Profit factor	1.11
Calmar (PnL / MaxDD)	0.96

Performance Metrics

Metric	Value
Total trades (n)	204
Win rate	42.6%
Sharpe (annualized)	1.99 [OOS]
Total P&L (backtest)	\$94.04
Avg P&L per trade	\$0.46
Validation gates passed	OOS PASS + beats BH

Caveats: Wider trail = fewer exits = larger holds. Marginal vs N50_ATR1.5.

See "Donchian Channel Trend Following (A1)" on the explanation pages.

#10 — C3_SOL_4h_vol_confirmed_Donchian

Tier: T3 Breakout

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	4h
Hold horizon	Variable (ATR trail)
Backtest window	~5.6y Binance SOL

Signal & Exit

Field	Value
Entry trigger	Close > prior 20-bar HIGH + 0.5*ATR(14) AND volume > 2x 20-bar avg -> LONG (mirror SHORT)
Exit rule	ATR(14) trailing stop, mult = 2.5

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$250
Max drawdown (on \$250 notional)	\$258.86
Max DD as % of notional	-103.5%
Profit factor	1.76
Calmar (PnL / MaxDD)	7.36

Performance Metrics

Metric	Value
Total trades (n)	211
Win rate	50.7%
Sharpe (annualized)	5.28 [OOS]
Total P&L (backtest)	\$1904.01
Avg P&L per trade	\$9.02
Total P&L (OOS)	\$969.83
Validation gates passed	5/6 (perm p=0.005, loses to SOL bull buy-hold)

Caveats: Loses to SOL bull-run buy-and-hold (\$1,904 vs \$21,145 in SOL bull). Useful for non-correlated PnL diversification, NOT to beat outright long-SOL exposure.

See "Volume-Confirmed Donchian Breakout (C3)" on the explanation pages.

#11 — E4_Cyclops_confidence_sizing_SOL_4h

Tier: T4 Composite

Market & Execution

Field	Value
Asset	SOL
Venue	Hyperliquid Perp
Signal timeframe	4h
Hold horizon	Variable (signal-flip)
Backtest window	~5.6y Binance SOL

Signal & Exit

Field	Value
Entry trigger	EMA stack score >= +4 (bull alignment) -> LONG; <= -4 -> SHORT; SIZE scales with Cyclops 3-a
Exit rule	Signal flip (stack score crosses zero)

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$250
Max drawdown (on \$250 notional)	\$102.81
Max DD as % of notional	-41.1%
Profit factor	1.46
Calmar (PnL / MaxDD)	2.78

Performance Metrics

Metric	Value
Total trades (n)	134
Win rate	56.7%
Sharpe (annualized)	1.98
Total P&L (backtest)	\$285.54
Avg P&L per trade	\$2.13
Avg funding paid per trade	\$0.060
Avg fee paid per trade	\$0.230
Validation gates passed	passes G1-G5 (G6 boot CI wide)

Caveats: Sizing layer on top of EMA-stack trend. Beats A2 baseline by +0.82 Sharpe on SOL 4h. Does NOT work well on BTC/SOL alone.

See "Cyclops 3-Axis as Confidence Multiplier (E4)" on the explanation pages.

#12 — F3_ETH_4h_rf_dist_bps_z1.5_momentum

Tier: T5 ML-derived

Market & Execution

Field	Value
Asset	ETH
Venue	Hyperliquid Perp
Signal timeframe	4h
Hold horizon	Variable (ATR SL/TP)
Backtest window	Walk-forward 4 windows on Binance ETH

Signal & Exit

Field	Value
Entry trigger	rolling 30d z-score of feature `rf_dist_bps` > +1.5 -> LONG (momentum follow); reverse SHORT for < -1.5
Exit rule	ATR SL/TP (sl_mult=1.0, tp_mult=2.5)

Capital & Risk

Field	Value
Notional per entry	\$250 (1x leverage)
Working capital required	\$250
Max drawdown (on \$250 notional)	not reported by D engine
Profit factor	n/a
Calmar (PnL / MaxDD)	n/a

Performance Metrics

Metric	Value
Total trades (n)	48
Win rate	46.4%
Sharpe (annualized)	1.16
Total P&L (backtest)	\$220.16
Avg P&L per trade	\$4.31
Validation gates passed	3/4 windows positive (not all 4)

Caveats: Single-feature rule derived from ML feature-importance. Marginal Sharpe. Not a full ML classifier (those lose to fees).

See "Single-Feature Z-Score Rule from ML Feature Importance (F3)" on the explanation pages.

Strategy Mechanisms — How Each Works

Basis Carry (D1) — Spot-Perp Mispricing Reversion

The Hyperliquid perpetual price and the underlying Binance spot price normally trade within a small band of each other. The exact 'fair' gap (basis) depends on the funding rate the perp is paying or receiving — if funding is +1 bp/hr, a 24-hour-forward perp price is expected to sit ~24 bps above spot (longs pay shorts that much).

Mispricing signal: at each hourly bar we compute:

```
basis_bps = (binance_spot_close - hl_perp_close) / hl_perp_close * 10000
expected_basis = (funding_rate_proxy) * (hold_hours / 8) * 10000
mispricing = basis_bps - expected_basis
```

We compute a rolling z-score of the mispricing over a configurable lookback window (7d, 14d, 30d, 60d, or 90d). When mispricing z-score exceeds the threshold:

- $z > +1.5$ (or +2.0, +2.5...) means HL perp is CHEAP relative to expected fair -> LONG HL perp
- $z < -1.5$ means HL perp is RICH -> SHORT HL perp

We hold for a fixed time horizon (24h, 48h, 72h, or 168h) and let funding accrue. Convergence toward fair pulls our P&L positive. Win rate is 55-95% depending on z-threshold (higher z = rarer signal = higher WR).

Why this works on Hyperliquid specifically: HL has a smaller, less efficient perp market than Binance. When sentiment piles into one side, the perp dislocates from spot. Most market participants don't have clean Binance+HL data side-by-side to spot this. The funding rate already prices in expected basis decay, so any deviation from that funding-implied basis is a real mispricing.

Expected-basis proxies tested:

- fund30davg: 30-day rolling mean of HL funding rate (smooth, slower)
- term_next: current funding rate as the implied term-structure (responsive to regime changes)
- zero_fund: assume funding is zero (pure spot-perp gap) — works at large z because funding is a small fraction of the basis at extremes

Direction verdict (D2): Funding-extreme CONTRARIAN beats funding-extreme MOMENTUM on all four HL coins tested (BTC, ETH, SOL, HYPE). When funding gets very positive, market is over-long; the marginal long is structurally weak. Fade it.

Donchian Channel Trend Following (A1)

Classical trend-following: a position is opened when price breaks above the highest high of the last N bars (long) or below the lowest low (short). The position is held with a trailing stop based on Average True Range (ATR) until the trail is hit.

Parameters tested: $N = \{20, 50, 100\}$; ATR multiplier for trail = $\{1.5, 2.0, 3.0\}$

Entry rule:

LONG: $\text{close}[t] > \max(\text{high}[t-N:t])$ (close breaks above N-bar high)
SHORT: $\text{close}[t] < \min(\text{low}[t-N:t])$

Exit rule:

For LONG: $\text{trail} = \text{running_max}(\text{high since entry}) - \text{ATR_mult} * \text{ATR}(14)$
Exit when $\text{low}[t] \leq \text{trail}$ (next bar open at the close of the breach bar)
Mirror logic for SHORT

The ETH 4h N=50 variant is the strongest because at 4h on ETH the bar size is large enough that the breakout is real (not noise) but small enough that breakouts happen frequently. ATR trail at 1.5x is tight enough to lock in moves before reversal but loose

enough to ride trends.

Why this works on crypto perp: Crypto is highly momentum-driven; sustained trends over hours-to-days are common. Funding cost on 4h holds is tiny (~1-3 bps round-trip). Trail prevents giving back accumulated gains. Beats buy-and-hold Sharpe on ETH by 5.6x because it goes flat during chop/drawdowns.

Volume-Confirmed Donchian Breakout (C3)

Same Donchian breakout as A1, but with TWO additional gates:

1. **ATR buffer on the breakout level:** close must exceed the N-bar high by at least $0.5 * ATR(14)$. This avoids triggering on tiny range expansions that are likely noise.

2. **Volume confirmation:** the bar must have volume $> 2x$ the 20-bar average volume. Real breakouts have institutional flow behind them; chop breakouts don't.

Why this works on SOL 4h: SOL has high realized volatility, so breakouts that ALSO have volume confirmation are rare but strong. The 5.3 OOS Sharpe is real; perm $p = 0.005$ confirms it's not random.

Important caveat: Over a SOL bull run (e.g. 2020-2021), simply being long SOL the whole time produces \$21k of PnL on \$250 — far better than the strategy's \$1,900. The strategy is valuable for DIVERSIFICATION (uncorrelated PnL during chop and bear regimes) not as a replacement for outright long-SOL exposure.

Cyclops 3-Axis as Confidence Multiplier (E4)

Cyclops is a multi-axis indicator coherence concept ported from Polymarket research. Three axes must align for a trade:

1. **Trend axis:** 20-bar linear regression slope sign + EMA stack alignment
2. **Levels axis:** proximity to and tested integrity of recent swing high/low pivots
3. **Momentum axis:** RSI(14) directional alignment AND no momentum exhaustion flag

The PERP-NATIVE adaptation is to use Cyclops not as a binary entry trigger (as in Polymarket) but as a POSITION-SIZING multiplier on top of a base strategy (here: EMA-stack-cross trend).

Sizing rule:

- All 3 axes aligned with trade direction -> 2x notional
- 2 of 3 axes aligned -> 1x notional
- Less than 2 -> skip the trade

This produces a +0.82 Sharpe lift over the unsized A2 (EMA-stack-only) baseline on SOL 4h. It's the textbook way to use a multi-indicator confluence in futures: as confidence, not as on/off filter.

The strategy works on SOL specifically — same sizing layer does NOT lift BTC or ETH meaningfully.

Single-Feature Z-Score Rule from ML Feature Importance (F3)

W3 (the meta-classifier wave) trained LightGBM/XGBoost on 100+ engineered features to predict next-bar direction. The full classifier doesn't generate enough edge to overcome HL's 12 bps round-trip cost (taker 4.5 bps x 2 + slippage 3 bps). However, permutation importance identified which individual features carry signal.

The F3 strategy uses just ONE feature (the strongest from the ML feature ranking) as a threshold rule:

feature = rf_dist_bps (distance from Range Filter band in basis points)

z_30d = rolling 30-day z-score of feature

LONG when $z > +1.5$ (with momentum bias)

SHORT when $z < -1.5$

Exit: ATR(14) SL at 1.0x ATR, TP at 2.5x ATR

Why this works while the full classifier doesn't: with one feature the rule is simple, captures the strongest signal, doesn't get noise from 99 other features, and stays robust across walk-forward windows. 3 of 4 walk-forward windows produced positive PnL.

Marginal Sharpe 1.16 — useful as a small position in a diversified book, not a primary deploy.

Pre-Deploy Checklist

1. **Refresh data:** HL canonical klines ended 2026-05-16; today is 2026-05-26. Pull the 10-day delta and re-run the carry signals before going live. Sharpe should remain positive on fresh data.
2. **Latency validation:** backtest assumes 50 ms. Verify against actual HL API round-trip from production environment. If latency > 200 ms, redo backtest with realistic figure.
3. **Slippage validation:** backtest assumes 3 bps slippage. Pull recent HL trade tape at \$250 notional, compute realized slippage, confirm 3 bps is conservative.
4. **Funding cap:** verify HL contract caps funding at 1.25 bps/hr — this is in the engine config; if HL changes the cap the strategy economics shift.
5. **Per-strategy notional scaling:** all backtests are \$250 notional 1x leverage. Scaling to \$1,000+ should be linear up to a point; monitor slippage as size grows.
6. **Drawdown circuit breaker:** auto-pause any sleeve that hits 60% of its backtested max drawdown. Re-evaluate the cause before un-pausing.
7. **Carry funding monitoring:** T1 carry strategies hold 24-168h with funding accrual. Max funding drag estimated ~10-30 bps per trade. Monitor real funding accrual vs backtested expectation closely.

Rejected Strategy Families (audit trail)

Mean Reversion (5 variants tested: RSI, BB, Z-score, SMS, VWAP)

5 / 90 cells profitable. Buy-and-hold dominates every variant. Crypto perp is momentum-driven, not mean-reverting on the timeframes tested.

ML Probability-Trading (RandomForest + LightGBM + XGBoost ensembles)

AUC edge of 2-4 ppt is real but smaller than HL's 12 bps round-trip cost. Net P&L; is negative on every cell. Use ML for feature selection only (see F3).

Hedged Spot-Perp Arbitrage (D3 — long HL perp + short Binance perp simultaneously)

All 27 cells negative. 2-venue fees + slippage = ~25 bps round-trip x 2 sides. Even when convergence is real (it is — corr near 0), fees eat the alpha.

Funding-Regime Composite (D4 — basis carry + funding-regime filter overlay)

High raw Sharpe (BTC 7.09) but fails G7 regime hold-out — collapses in at least one regime. Hidden concentration risk; do NOT deploy raw.

Markov regime router (E1) and session-switch (E3)

Both drag relative to best component. Forcing mean-reversion in sideways regimes destroys good trend trades.

All 5m and 15m timeframes for non-carry strategies

HL fees crush sub-bar edges. Only carry strategies (which need long enough horizons for convergence) can amortize fees.